DSA REFERENCE

Fast and Slow Pointers Failure Modes

Common wrong turns, debugging cues, and complexity traps.

Move two traversals at different speeds so their relative motion reveals cycles, midpoints, or phase boundaries.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / FAST-SLOW-POINTERS

DSA

Fast and Slow Pointers Failure Modes

CORE IDEA

Move two traversals at different speeds so their relative motion reveals cycles, midpoints, or phase boundaries.

Recognition cues

+ linked-list cycle
+ middle node
+ repeated state transition
+ constant-space sequence detection

Implementation sequence

+ guard every multi-step dereference
+ explain relative speed
+ separate detection from entry location
+ test odd and even lengths

Invariant

Both pointers follow the same deterministic next relation; only their speed or restart position differs.

Worked reasoning

After slow and fast meet in a cycle, reset one pointer to head and move both one step to find the entry.

Complexity

O(n) time and O(1) space for linked structures or deterministic state machines.

Failure modes

- [ ] unsafe fast.next.next access
- [ ] returning the meeting point as the entry
- [ ] misstating even-length midpoint
- [ ] using it when transitions are nondeterministic

Practice ladder

Fast and Slow Pointers Failure Modes: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Linked List Cycle**	State the cue, invariant, edge case, and proof before implementation.
2	**Linked List Cycle II**	State the cue, invariant, edge case, and proof before implementation.
3	**Middle of the Linked List**	State the cue, invariant, edge case, and proof before implementation.
4	**Happy Number**	State the cue, invariant, edge case, and proof before implementation.
5	**Find the Duplicate Number**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS

REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.